

5. Execution Flow & Daily/Weekly QA Operating Model – Plan Conversion QA Program

1. Purpose

This page defines **how QA will execute the project day-to-day and week-to-week**, including:

- Daily smoke routines
- Weekly regression cycles
- Automation entry point logic
- File feed & daemon monitoring
- Kofax execution flow
- CSR/Member execution order
- Environment readiness checks
- Escalation boundaries

This is the operational model the entire QA team must follow across Stage1 and QC4.

2. High-Level Execution Framework

The QA execution framework is built on three pillars:

1. API-First Testing

Ensure backend correctness first, then UI.

2. Controlled Data Lifecycle

Use deterministic test data; avoid stale or unknown accounts.

3. Incremental Regression

Run focused smoke daily full regression weekly.

3. Daily Execution Flow (Critical)

The daily routine is non-negotiable.

This stabilizes the environments and reduces rework.

3.1 Daily Smoke Checklist (Automation + Manual)

Team completes this before ANY testing:

A. Environment Ready

- Verify Stage1 availability
- Verify QC4 availability
- Validate SSO handshake manually
- Validate IDP health (no 5xx errors)
- Validate CSR login works
- Validate Kofax endpoints reachable
- Validate file drop directories accessible

B. Metadata Health Checks

- Run metadata GET call
- Verify PlanID, funds, tranche mapping
- Validate contribution/withdrawal rules
- Validate SSO flags
- Validate UGIFT flags

C. API Smoke Execution

Run fast API-only smoke (~30 tests):

- Account POST/GET

- Contribution POST
- Withdrawal POST
- Transfer
- Allocation GET
- Plan GET
- Metadata GET

If API smoke fails stop all UI testing.

D. DB Smoke

- Validate 1 test user
- Validate 1 transaction
- Validate 1 Kofax ingest from previous day
- Validate 1 file feed run (Price/Recon)

E. File Feed Monitoring

Check logs for:

- Price
- Performance
- Recon
- Max Contribution
- Account Sync
- Flywire

Any failure block regression.

3.2 Daily Execution Order

Step 1 – API Smoke

If fails stop.

Step 2 – CSR Light Smoke

- Login
- Overview
- Balance
- Recent transactions

Step 3 – Member Smoke (manual)

- SSO redirect
- Profile
- Contribution

Step 4 – Kofax Smoke

- 1 enrollment file ingest
- Validate DB & CSR visibility

Step 5 – File Feeds Smoke

- Validate price/performance/Recon

Step 6 – Defect triage & escalations

4. Weekly Execution Flow (Full Regression)

Full regression happens weekly to detect drift early.

4.1 Weekly Regression Checklist

A. Pre-Regression Steps

- DB snapshot taken

- Metadata validated
- File feeds paused (to prevent overwrites)
- Test data regenerated via API
- CSR user permissions validated
- SSO accounts validated manually

B. Regression Execution Order

1. API Regression (Full)

Fastest. Most reliable.

If API breaks no UI.

2. CSR Regression (Full Automation)

~800–1,200 tests depending on scope.

3. Member Regression (Hybrid)

- Manual login + initial flows
- Automation from post-SSO
- Validate: contribution, withdrawal, transfers, image uploads

4. Backoffice Regression

Feed DB CSR integrity:

- Price
- Performance
- Recon
- Flywire
- ACH
- Gifting reversals
- Positive Pay

5. Kofax Full Suite

All feed categories.

6. Conversion Health Suite

Final after everything else.

4.2 Weekly Regression Exit Criteria

Area	Criteria
API	95%+ pass (env failures excluded)
CSR	90%+ pass, no critical failures
Member	All critical flows pass manually + automation
Backoffice	All feeds consumed + DB mapping verified
Kofax	No schema mismatches
Conversion suite	100% pass

5. Execution Flow by Functional Area

5.1 API Execution Flow

1. Create test data via Account POST
2. Run functional API tests
3. Run negative cases
4. Validate DB
5. Validate CSR screens
6. Produce JSON health report

APIs drive the entire testing strategy.

5.2 CSR Execution Flow

CSR automation runs in this order:

1. Login

2. Overview & balance
3. Transaction flows
4. Maintenance flows
5. Member actions
6. Profile & documents
7. Beneficiary & image flows
8. Fee flows
9. Authorized agents
10. Trusted contact
11. Rep/Advisor
12. Bank info
13. Payroll

CSR is the highest automation ROI.

5.3 Member Portal Execution Flow

Since SSO cannot be automated:

1. Manual SSO login
2. Automation picks up after redirect
3. Execute:
 - Contribution
 - Recurring contributions
 - Rollover
 - Transfer bene
 - Change investment
 - Profile
 - Image upload
4. Validate DB
5. Validate CSR

5.4 Backoffice Execution Flow

1. Drop test files
2. Trigger daemon (manual or cron)
3. Monitor logs
4. Validate DB inserts
5. Validate CSR views
6. Validate Member views
7. Validate reconciliation data

All file feeds follow this flow.

5.5 Kofax Execution Flow

1. Inject XML into drop folder
2. Validate ingestion logs
3. Validate routing
4. Validate DB staging
5. Validate CSR visibility
6. Validate error folder (if applicable)
7. Validate conversion mapping

6. Environment Readiness Checklist

A. Weekly Requirements

- Metadata loaded
- Tranche/fund mapping active
- All APIs enabled
- Kofax endpoints functioning
- File feeds linked to correct directory
- Daemons scheduled
- SSO handshake stable

B. Daily Requirements

- API smoke successful
- CSR login working
- Member login working manually

- No major backend outages
- File feed logs clean

7. Escalation Rules

Escalate to Backend When:

- Metadata mismatches
- Fund/tranche mapping issues
- API returns incorrect data

Escalate to DevOps When:

- File feeds fail
- Jobs do not trigger
- Directory permissions fail

Escalate to Imaging When:

- Kofax ingestion problems
- XSD/schema mismatches
- Unexpected errors in routing

Escalate to WhiteCap When:

- SSO breaks
- Tokens expire
- Redirect fails

Escalate to Leadership When:

- Blocking defects across systems
- Metadata freeze ignored
- Kofax schema changes late
- Backoffice delays

8. Roles & Responsibilities

Role	Responsibilities
Senior SDET	API framework, CSR updates, DB validation, backoffice file automation
Mid-Level SDET	CSR execution, Kofax automation, file watchers, test data creation
Manual QA	SSO flows, MFA, UI validation, content, mobile, aggregator
DevOps	File paths, job triggers, feed schedule, access
Backend	Metadata, tranches, fund mapping, contribution limits
Imaging/Kofax	Feeds, XSD, ingestion routing
WhiteCap	SSO tokens, redirect flows
You (QA Lead)	Architecture, oversight, planning, reporting
Pramod (QA Architect)	SME, triage, technical validation

9. Final Notes

To keep drift low and execution predictable:

- Automation **always starts with API**
- UI runs are never first
- SSO login is manual ONLY
- DB validation is mandatory for every critical flow
- Weekly snapshots guarantee repeatability
- Daily smoke ensures early-break detection

- Kofax must be executed in controlled windows
- File feeds must be paused during regression

This execution model ensures **9-week delivery** is realistic.